Proof

(i) Let $x = \sin\theta$. Thus, $\theta = \sin^{-1} x$.

Now, $3x - 4x^3 = 3\sin\theta - 4\sin^3\theta = \sin 3\theta$.

Thus, $\sin^{-1}(3x - 4x^3) = 3\theta = 3\sin^{-1} x$.

The other result is proved in a similar way.

Remark

(i) $$\sin^{-1}(\cos x) = \begin{cases} \frac{\pi}{2} - x, & \text{if } x \in [0, \pi] \\ \frac{\pi}{2} - y, & \text{if } x \notin [0, \pi] \text{ and } \cos x = \cos y, y \in [0, \pi] \end{cases}$$

(ii) $$\cos^{-1}(\sin x) = \begin{cases} \frac{\pi}{2} - x, & \text{if } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \frac{\pi}{2} - y, & \text{if } x \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ and } \sin x = \sin y,\ y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \end{cases}$$

Example 4.16

Prove that $\frac{\pi}{2} \le \sin^{-1} x + 2\cos^{-1} x \le \frac{3\pi}{2}$.

Solution

$$\sin^{-1} x + 2\cos^{-1} x = \sin^{-1} x + \cos^{-1} x + \cos^{-1} x = \frac{\pi}{2} + \cos^{-1} x$$

We know that $0 \le \cos^{-1} x \le \pi$. Thus, $\frac{\pi}{2} + 0 \le \cos^{-1} x + \frac{\pi}{2} \le \pi + \frac{\pi}{2}$.

Thus, $\frac{\pi}{2} \le \sin^{-1} x + 2\cos^{-1} x \le \frac{3\pi}{2}$.

Example 4.17

Simplify (i) $\cos^{-1}\left(\cos\left(\frac{13\pi}{3}\right)\right)$ (ii) $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$

(iii) $\sec^{-1}\left(\sec\left(\frac{5\pi}{3}\right)\right)$ (iv) $\sin^{-1}[\sin 10]$

Solution

(i) $\cos^{-1}\left(\cos\left(\frac{13\pi}{3}\right)\right)$. The range of principal values of $\cos^{-1} x$ is $[0, \pi]$.

Since $\frac{13\pi}{3} \notin [0, \pi]$, we write $\frac{13\pi}{3}$ as $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$, where $\frac{\pi}{3} \in [0, \pi]$.

Now, $\cos\left(\frac{13\pi}{3}\right) = \cos\left(4\pi + \frac{\pi}{3}\right) = \cos\frac{\pi}{3}$.

Thus, $\cos^{-1}\left(\cos\left(\frac{13\pi}{3}\right)\right) = \cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3}$, since $\frac{\pi}{3} \in [0, \pi]$.

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